

Musterlösung Serie 4

LINEAR SYSTEMS, ROW REDUCTION

The first exercise is repeated from the previous exercise sheet, since the necessary material wasn't yet covered.

1. Linear system. Fix $b_1, b_2, b_3 \in \mathbb{R}$. Determine when the following linear system of equations has a solution and describe its set of solutions $S \subseteq \mathbb{R}^3$ when it does

$$\begin{pmatrix} 0 & 7 & -2 \\ -1 & 2 & -1/2 \\ 4 & -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}.$$

Solution: We write the augmented matrix for this system

$$\begin{aligned} \left(\begin{array}{ccc|c} 0 & 7 & -2 & b_1 \\ -1 & 2 & -1/2 & b_2 \\ 4 & -1 & 0 & b_3 \end{array} \right) &\rightsquigarrow \left(\begin{array}{ccc|c} -1 & 2 & -1/2 & b_2 \\ 4 & -1 & 0 & b_3 \\ 0 & 7 & -2 & b_1 \end{array} \right) \rightsquigarrow \left(\begin{array}{ccc|c} 1 & -2 & 1/2 & -b_2 \\ 4 & -1 & 0 & b_3 \\ 0 & 7 & -2 & b_1 \end{array} \right) \\ &\rightsquigarrow \left(\begin{array}{ccc|c} 1 & -2 & 1/2 & -b_2 \\ 0 & 7 & -2 & b_3 + 4b_2 \\ 0 & 7 & -2 & b_1 \end{array} \right) \rightsquigarrow \left(\begin{array}{ccc|c} 1 & -2 & 1/2 & -b_2 \\ 0 & 7 & -2 & b_3 + 4b_2 \\ 0 & 0 & 0 & b_1 - b_3 - 4b_2 \end{array} \right) \end{aligned}$$

So, the equivalent system is

$$\begin{cases} x - 2y + \frac{1}{2}z &= -b_2 \\ 7y - 2z &= b_3 + 4b_2 \\ 0 &= b_1 - b_3 - 4b_2 \end{cases}$$

If $b_1 - b_3 - 4b_2 \neq 0$, then the system does not have any solutions. If we do have $b_1 - b_3 - 4b_2 = 0$, then the set of solutions is

$$S = \left\{ (x, y, z) \in \mathbb{R}^3 \mid x = \frac{1}{14}(z + 4b_3 + 2b_2), y = \frac{1}{7}(2z + b_3 + 4b_2) \right\}.$$

2. Linear system. For which values of $a \in \mathbb{R}$ and $b \in \mathbb{R}$ does the following system admit solutions $(x, y, z) \in \mathbb{R}^3$?

$$\begin{aligned}x + 2y - 3z &= 10 \\3x + 7y - 9z &= b \\-3x - 5y + az &= b \\2x + 4y - 3z &= 5\end{aligned}$$

For all such values of a and b give all the solutions.

Solution: We write the augmented matrix of the system and reduce it

$$\begin{aligned}\left(\begin{array}{ccc|c}1 & 2 & -3 & 10 \\3 & 7 & -9 & b \\-3 & -5 & a & b \\2 & 4 & -3 & 5\end{array}\right) &\rightsquigarrow \left(\begin{array}{ccc|c}1 & 2 & -3 & 10 \\0 & 1 & 0 & b-30 \\0 & 1 & a-9 & b+30 \\0 & 0 & 3 & -15\end{array}\right) &\rightsquigarrow \left(\begin{array}{ccc|c}1 & 0 & -3 & 70-2b \\0 & 1 & 0 & b-30 \\0 & 0 & a-9 & 60 \\0 & 0 & 3 & -15\end{array}\right) \\&\rightsquigarrow \left(\begin{array}{ccc|c}1 & 0 & -3 & 70-2b \\0 & 1 & 0 & b-30 \\0 & 0 & a-9 & 60 \\0 & 0 & 1 & -5\end{array}\right) &\rightsquigarrow \left(\begin{array}{ccc|c}1 & 0 & 0 & 55-2b \\0 & 1 & 0 & b-30 \\0 & 0 & 1 & -5 \\0 & 0 & 0 & 60-5(9-a)\end{array}\right).\end{aligned}$$

Hence, this system has solutions when

$$0 = 60 - 5(9 - a) \Leftrightarrow a = -3$$

and for all $b \in \mathbb{R}$. For any pair $(a, b) = (-3, b)$, the solution is

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 55 - 2b \\ b - 30 \\ -5 \end{pmatrix}.$$

3. Linear system.

Consider the following two systems of equations.

$$\begin{cases} x + y + z &= 6 \\ x + 2y + 2z &= 11 \\ 2x + 3y - 4z &= 3 \end{cases}$$

and

$$\begin{cases} x + y + z &= 7 \\ x + 2y + 2z &= 10 \\ 2x + 3y - 4z &= 3 \end{cases}$$

Solve both systems of equations simultaneously using row-reduction to an appropriate 3×5 extended matrix.

Solution: We write

$$\begin{aligned} \left(\begin{array}{ccc|cc} 1 & 1 & 1 & 6 & 7 \\ 1 & 2 & 2 & 11 & 10 \\ 2 & 3 & -4 & 3 & 3 \end{array} \right) &\rightsquigarrow \left(\begin{array}{ccc|cc} 1 & 1 & 1 & 6 & 7 \\ 0 & 1 & 1 & 5 & 3 \\ 0 & 1 & -6 & -9 & -11 \end{array} \right) &\rightsquigarrow \left(\begin{array}{ccc|cc} 1 & 0 & 0 & 1 & 4 \\ 0 & 1 & 1 & 5 & 3 \\ 0 & 0 & -7 & -14 & -14 \end{array} \right) \\ &\rightsquigarrow \left(\begin{array}{ccc|cc} 1 & 0 & 0 & 1 & 4 \\ 0 & 1 & 1 & 5 & 3 \\ 0 & 0 & 1 & 2 & 2 \end{array} \right) &\rightsquigarrow \left(\begin{array}{ccc|cc} 1 & 0 & 0 & 1 & 4 \\ 0 & 1 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 & 2 \end{array} \right) \end{aligned}$$

Hence the solution to the first system is

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

and the solution to the second system is

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix}.$$

4. Row reduction. You learnt in class that if the augmented matrices of two linear systems of equations are equivalent, they have the same solutions. Is the converse true? If yes, prove it, and if not, provide a counterexample.

Solution: The converse is not true if we consider systems that do not admit any solutions. For example, consider the systems

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Both systems do not admit any solutions. However, the row reduction of

$$\left(\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & 1 & 1 \end{array} \right) \quad \text{is} \quad \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right)$$

while the row reduction of

$$\left(\begin{array}{cc|c} 1 & 2 & 0 \\ 1 & 2 & 1 \end{array} \right) \quad \text{is} \quad \left(\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 0 & 1 \end{array} \right).$$

There is no elementary row operation taking one row reduced matrix to the other, hence the two systems aren't equivalent.

5. Row reduction. Let A be a 2-times-2 real matrix and let $b \in \mathbb{R}^2$ such that the system of equations $Ax = b$ has no solutions. Show that one row of the matrix A is a multiple of the other.

Solution: Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad b = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

We show that if no row of A is a multiple of the other, then there exist a solution for all $\alpha, \beta \in \mathbb{R}$. First assume that $a \neq 0$. We have

$$\left(\begin{array}{cc|c} a & b & \alpha \\ c & d & \beta \end{array} \right) \rightsquigarrow \left(\begin{array}{cc|c} a & b & \alpha \\ 0 & d - \frac{c}{a}b & \beta - \frac{c}{a}\alpha \end{array} \right) \rightsquigarrow \left(\begin{array}{cc|c} 1 & \frac{b}{a} & \frac{\alpha}{a} \\ 0 & 1 & (\beta - \frac{c}{a}\alpha)/(d - \frac{c}{a}b) \end{array} \right)$$

Note that the last step is allowed because $\frac{c}{a} \neq \frac{d}{b}$ by assumption. The reduced system clearly admits a solution.

Now assume that $a = 0$. If $c \neq 0$ we use an elementary row operation to exchange the two rows and proceed as in the case $a \neq 0$. If $c = 0$ then either $b \neq 0$ or $d \neq 0$. Otherwise one of the two rows vanishes, which cannot happen since we assumed that no row was a multiple of the other. Now, if $b \neq 0$, the second row is a multiple of the first one with factor d/b . If $d \neq 0$, the same holds by switching the roles of d and b .

This shows that if the two rows of A are not multiple of one another then the system admits a solution for all $\alpha, \beta \in \mathbb{R}$ and hence proves the contrapositive.